

Phylogenetic Analysis Using NCBI and Galaxy

GROUP - Chiroptera
BIO 300 - Cell & Molecular Biology

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July 31, 2026

Introduction

Cytochrome c oxidase subunit I (COI or COX1) is a mitochondrial gene that encodes a subunit of cytochrome c oxidase, an essential enzyme involved in cellular respiration and ATP production. Because the COI gene is present in most animal species and contains both conserved and variable regions, it has become the standard molecular marker for DNA barcoding, species identification, and phylogenetic analysis. Conserved regions enable the alignment of sequences from different species, whereas variable regions provide sufficient genetic variation to distinguish closely related taxa and infer their evolutionary relationships (Hebert et al., 2003)

In this study, mitochondrial COI sequences retrieved from the National Center for Biotechnology Information (NCBI) were aligned using MAFFT in Galaxy to reconstruct the phylogenetic relationships among selected order Chiroptera, with *Mus musculus* included as the outgroup. The resulting phylogenetic tree was used to examine evolutionary relationships among the selected taxa and compare them with published molecular phylogenies.

Table 1. Summary of mitochondrial COI sequence data, accession numbers, and sequence lengths for selected Chiroptera species and the *Mus musculus* outgroup.

Species	Common name	NCBI accession number	Sequence Length	Record Title	Focal taxon / Outgroup
<i>Mus musculus</i>	House mouse	AB444046.1	702 bp	<i>Mus musculus</i> mitochondrial COI gene for cytochrome c oxidase subunit I, partial cds	Outgroup
<i>Hipposideros diadema</i>	Diadem leaf-nosed bat	HM914927.1	658 bp	<i>Hipposideros diadema</i> voucher ZMMU SVK 136-09 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal Taxon
<i>Rhinolophus arcuatus</i>	Arcuate horseshoe bat	MH014038.1	658 bp	<i>Rhinolophus arcuatus</i> voucher MRMD2465 cytochrome oxidase subunit 1 (COI) gene,	Focal Taxon

				partial cds; mitochondrial	
<i>Pteropus vampyrus</i>	Large flying fox	MK415810.1	627 bp	<i>Pteropus vampyrus</i> isolate PG-M3/COI/AJK/2018 cytochrome c oxidase subunit I (COI) gene, partial cds; mitochondrial	Focal Taxon
<i>Rousettus amplexicaudatus</i>	Geoffroy's rousette	MK585723.1	658 bp	<i>Rousettus amplexicaudatus</i> voucher MRMD2092 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal Taxon
<i>Miniopterus schreibersii</i>	Common bent-wing bat	MT407283.1	658 bp	<i>Miniopterus schreibersii</i> voucher MB14 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal Taxon
<i>Cynopterus brachyotis</i>	Lesser short-nosed fruit bat	MK585802.1	616 bp	<i>Cynopterus brachyotis</i> voucher TRB1 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial	Focal Taxon

Methods

COI gene sequences of the selected bat species - were retrieved from the NCBI Nucleotide database, together with *Mus musculus* as the outgroup. All sequences were downloaded in FASTA format and imported into Galaxy (usegalaxy.org) for analysis. Multiple sequence alignment was performed to identify homologous nucleotide positions

and assess genetic similarities among the sequences. A consensus sequence was subsequently generated to summarize the conserved nucleotide sites across the aligned COI sequences. The aligned sequences were then used to construct a phylogenetic tree using FastTree with the nucleotide sequence option. The resulting tree was visualized in Galaxy, and *Mus musculus* was included as the outgroup to aid in the interpretation of the evolutionary relationships among the selected bat species.

The figures below illustrate the workflow used to generate the consensus sequence and construct the phylogenetic tree from the COI sequences:

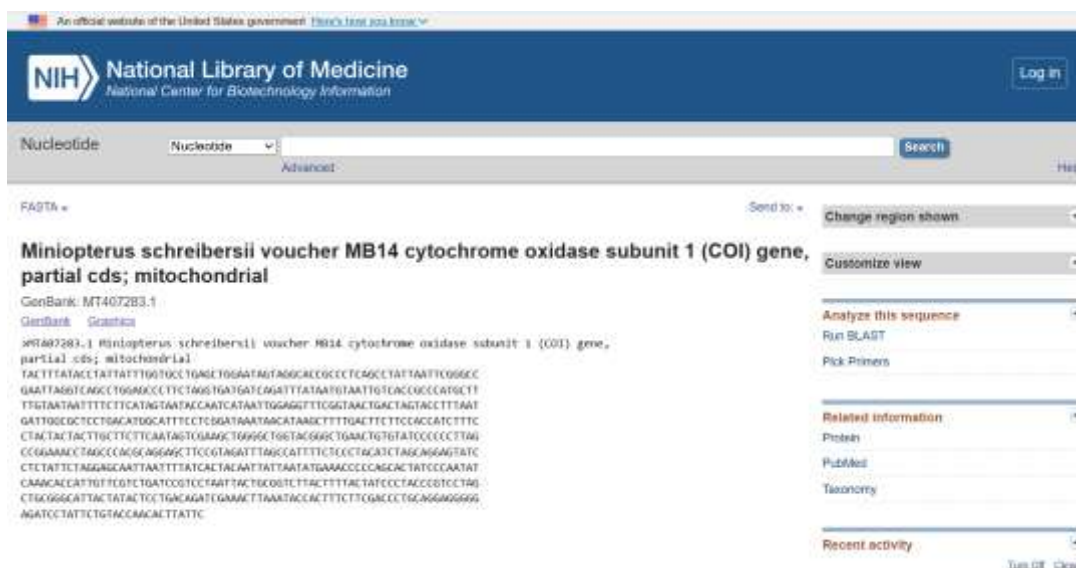


Figure 1. NCBI Nucleotide COI/COX1 sequences were searched by scientific name, verified for correct region and length, and saved as FASTA files.

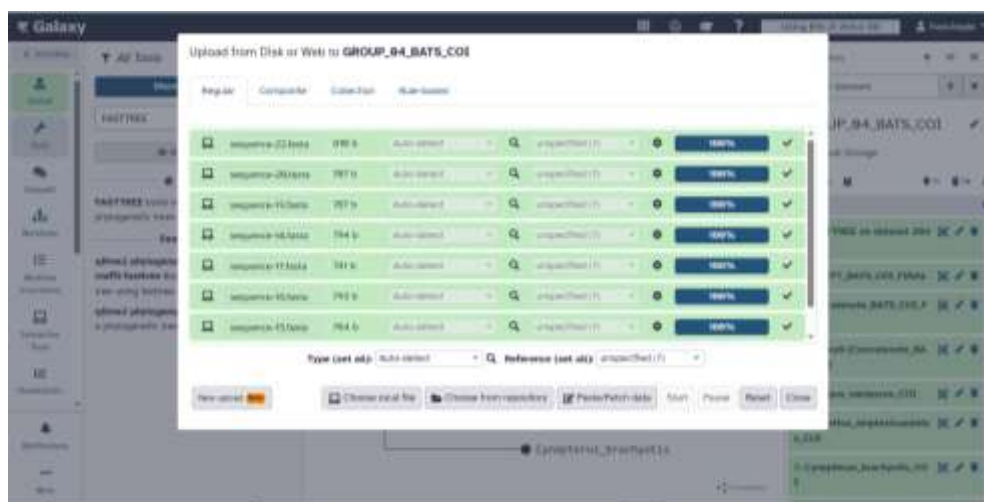


Figure 2. The downloaded COI sequences were uploaded into the Galaxy history

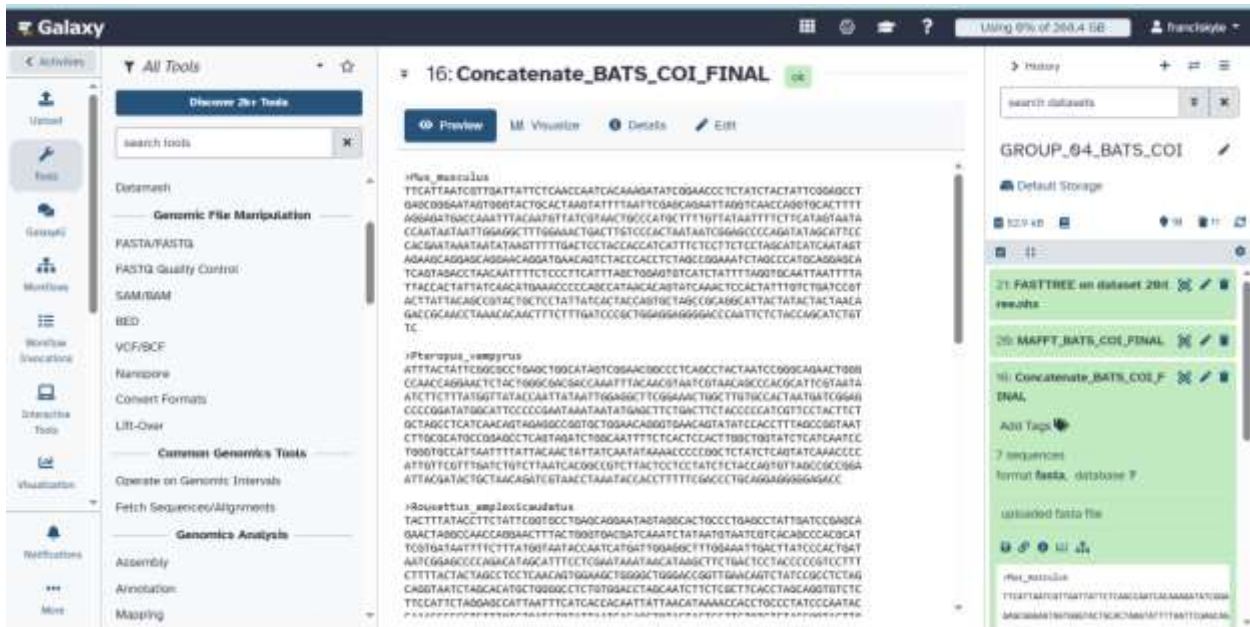


Figure 3. Concatenating individual COI FASTA records into a single dataset.

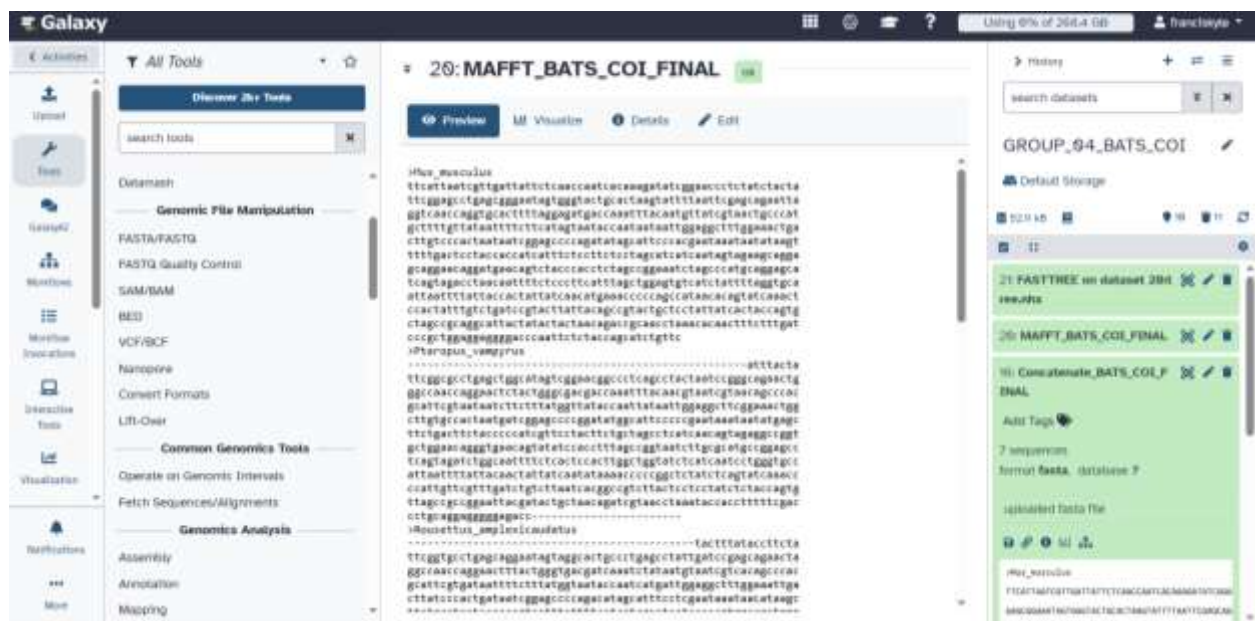


Figure 4. Running the MAFFT multiple sequence alignment tool in Galaxy.

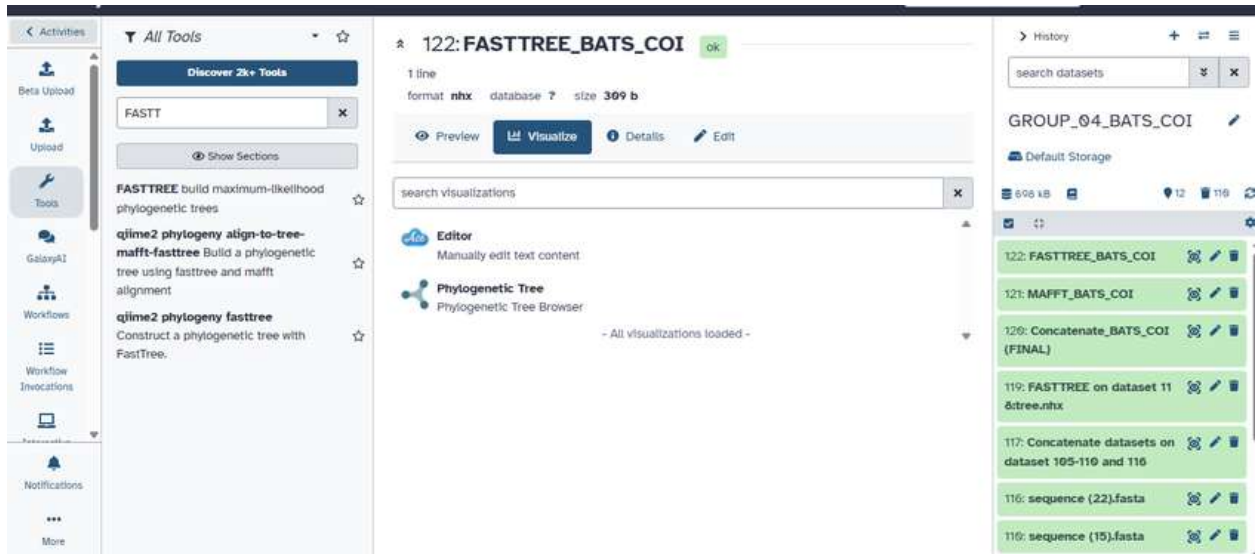


Figure 5. Construction of the phylogenetic tree using FastTree in Galaxy.

Summary of Conserved and Variable Sites

The multiple sequence alignment and consensus sequence analysis showed that the mitochondrial COI gene contains both conserved and variable nucleotide sites among the bat species. Most nucleotide positions were conserved, indicating that these regions have remained stable throughout evolution due to their essential role in cellular respiration. These conserved sites contribute to the similar genetic characteristics shared by the species.

In contrast, several variable sites were observed where nucleotide differences occurred among the sequences. These variations provide the genetic information needed to distinguish species and infer their evolutionary relationships. The combination of highly conserved and variable regions demonstrates that the COI gene is well suited for phylogenetic analysis, as it maintains functional conservation while containing sufficient sequence variation to resolve relationships among closely related bat species.

Overall, the conserved regions allowed the COI sequences to be aligned accurately, while the variable regions provided enough genetic differences to distinguish the bat species. This combination makes the COI gene a reliable molecular marker for species identification and phylogenetic analysis.



Figure 6. Consensus COI Sequence Generated from the Multiple Sequence Alignment of chosen Chiroptera Species

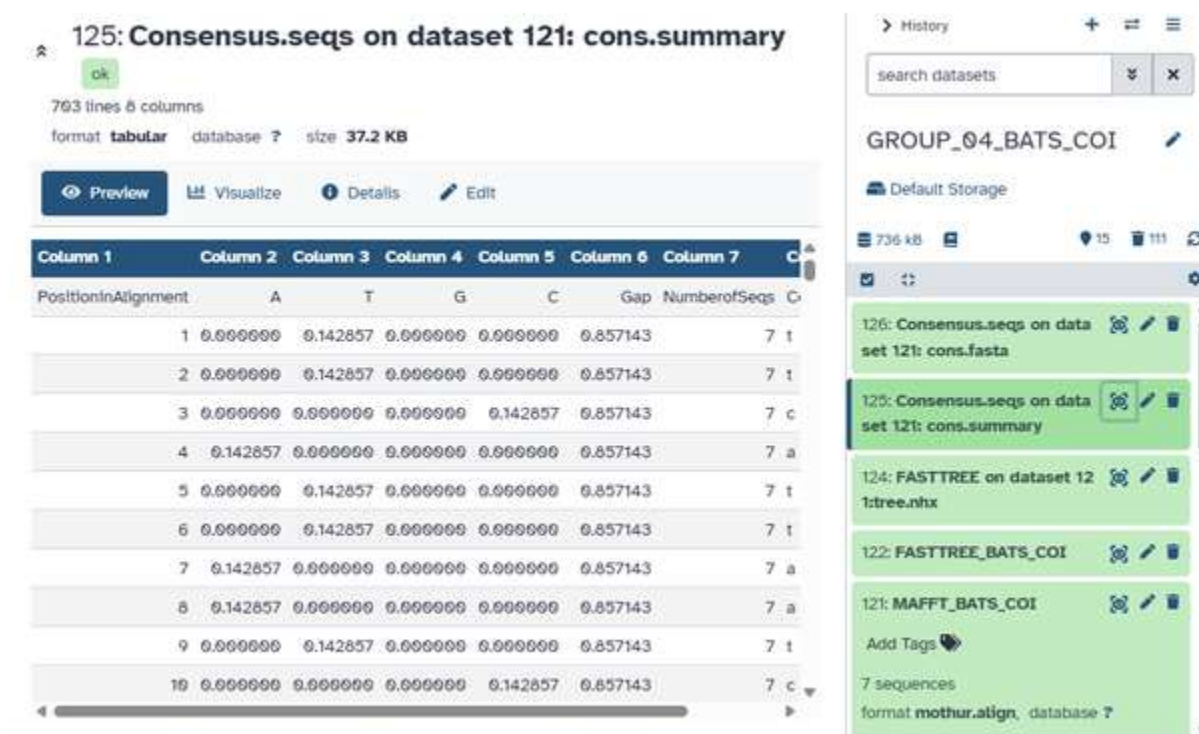


Figure 7. Consensus Sequence Summary Showing Conserved and Variable Sites

Comparison with Published Studies

The phylogenetic tree generated in this study is generally consistent with published molecular studies on bat evolution. The tree clearly separates the fruit bats (*Pteropus vampyrus*, *Rousettus amplexicaudatus*, and *Cynopterus brachyotis*) from the echolocating bats (*Miniopterus schreibersii*, *Hipposideros diadema*, and *Rhinolophus arcuatus*), supporting the major bat lineages described by Amador et al. (2018). In addition, the close relationships observed between *Hipposideros diadema* and *Rhinolophus arcuatus*, and between *Rousettus amplexicaudatus* and *Cynopterus brachyotis*, are consistent with previously reported molecular classifications of these species. The placement of *Mus musculus* outside all bat species also supports the accuracy of the phylogenetic tree. Since *Mus musculus* belongs to Order Rodentia, which is evolutionarily distinct from Chiroptera, it served as an appropriate outgroup for rooting the tree and interpreting the evolutionary relationships among the selected bat species (Mindell et al., 1991; Hao et al., 2024).

Although the overall topology agrees with published studies, slight differences in branching patterns are expected because this analysis was based only on the mitochondrial COI gene. In contrast, Shi and Rabosky (2015) used larger molecular datasets, including multiple genetic markers, which provide higher phylogenetic resolution and stronger statistical support. Despite this limitation, the present findings demonstrate that the COI gene is an effective molecular marker for species identification and for reconstructing the general evolutionary relationships among bats.

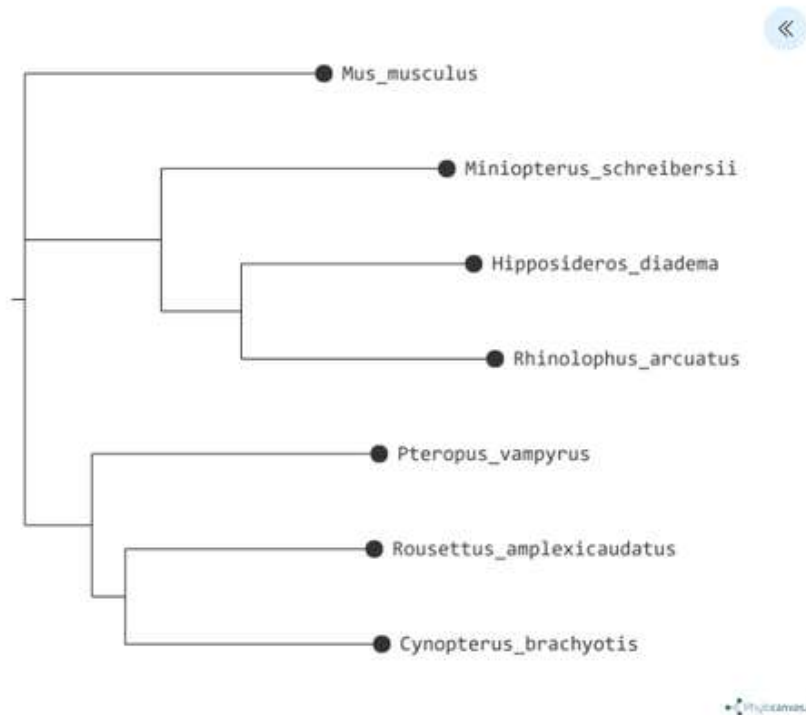


Figure 8. Phylogenetic tree of the selected Chiroptera species based on mitochondrial COI sequences generated using FastTree.

References

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